OEE Tracker

Step-by-step guide: track Overall Equipment Effectiveness by machine and shift

What it does

You log one row per machine per shift. The workbook calculates Availability, Performance, Quality and OEE for every row, then summarizes each machine on the **Dashboard** for any date range you choose, with a chart and a comparison to your target.

Yellow cells: type here.	Grey cells: calculated. Don't type over them.

The sheets

Sheet	What it's for
How to use	A one-page summary.
Setup	Your machines and each one's ideal cycle time. Up to 20 machines.
Production Log	One row per machine per shift. Up to 500 rows.
Dashboard	Totals and OEE by machine for a date range, plus a chart.

Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

Step 1: Set up your machines

1. Go to the **Setup** sheet.
2. In **Machine**, type each machine's name once, one per row (up to 20). Use the names your team already uses.
3. In **Ideal cycle time (seconds per part)**, enter the fastest time the machine can sustainably make one good part. For a line, use the time per finished unit at rated speed. Example: a machine rated at 300 parts per hour has an ideal cycle time of 3600 / 300 = **12 seconds**.
4. Type each machine only once. If a name is repeated, **Check** on the repeat says **Duplicate: only the first row is used**; delete the repeat.
5. Clear the three EXAMPLE machines when you no longer need them (yellow cells only).

Getting the ideal cycle time right matters most: if it's too slow, Performance can show over 100% and the Check column will flag it.

Step 2: Log each shift

1. Go to the **Production Log** sheet.
2. Fill in one row per machine per shift. The yellow columns are: **Date**, **Shift** (any label, e.g. Day or 1st), **Machine** (pick from the drop-down), **Shift length (min)**, **Planned stops (min)**, **Unplanned downtime (min)**, **Total parts** and **Rejected parts**. Shift length and total parts are required; blank planned stops, downtime or rejects count as 0.
3. **Planned stops** are times the machine isn't expected to run, like breaks and scheduled meetings. **Unplanned downtime** is every other minute it was stopped: breakdowns, changeovers, waiting for material.
4. The grey columns fill in by themselves, including **OEE** (coloured green, amber or red) and **Check**.
5. Look at **Check** on each new row. It should say **OK**. If not, see the table below. Rows with an error are left out of the Dashboard until you fix them. The "performance above 100%" warning is not an error: those rows still count.
6. The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.

Production Log

One row per machine per shift. Fill in the yellow columns; grey columns calculate.

Date	Shift	Machine	Shift length (min)	Planned stops (min)	Unplanned downtime (min)	Total parts	Rejected parts	Planned production time (min)	Run time (min)	Ideal cycle time (sec)	Ideal run time (min)	Good parts	Availability	Performance	Quality	OEE	Check	Ideal time for good parts (min)	Row used in totals
2026-09-01	Day	Press 01	480	40	35	1900	22	440	405	12.0	380.0	1878	92.0%	93.8%	98.8%	85.4%	OK	375.6	TRUE
2026-09-01	Day	CNC Lathe 02	480	40	60	210	4	440	380	95.0	332.5	206	86.4%	87.5%	98.1%	74.1%	OK	326.2	TRUE
2026-09-01	Day	Packaging Line A	480	40	25	9200	150	440	415	2.5	383.3	9050	94.3%	92.4%	98.4%	85.7%	OK	377.1	TRUE
2026-09-02	Day	Press 01	480	40	90	1500	30	440	350	12.0	300.0	1470	79.5%	85.7%	98.0%	66.8%	OK	294.0	TRUE
2026-09-02	Day	CNC Lathe 02	480	40	20	240	2	440	420	95.0	380.0	238	95.5%	90.5%	99.2%	85.6%	OK	376.8	TRUE
2026-09-02	Day	Packaging Line A	480	40	55	8400	95	440	385	2.5	350.0	8305	87.5%	90.9%	98.9%	78.6%	OK	346.0	TRUE
																			FALSE
																			FALSE

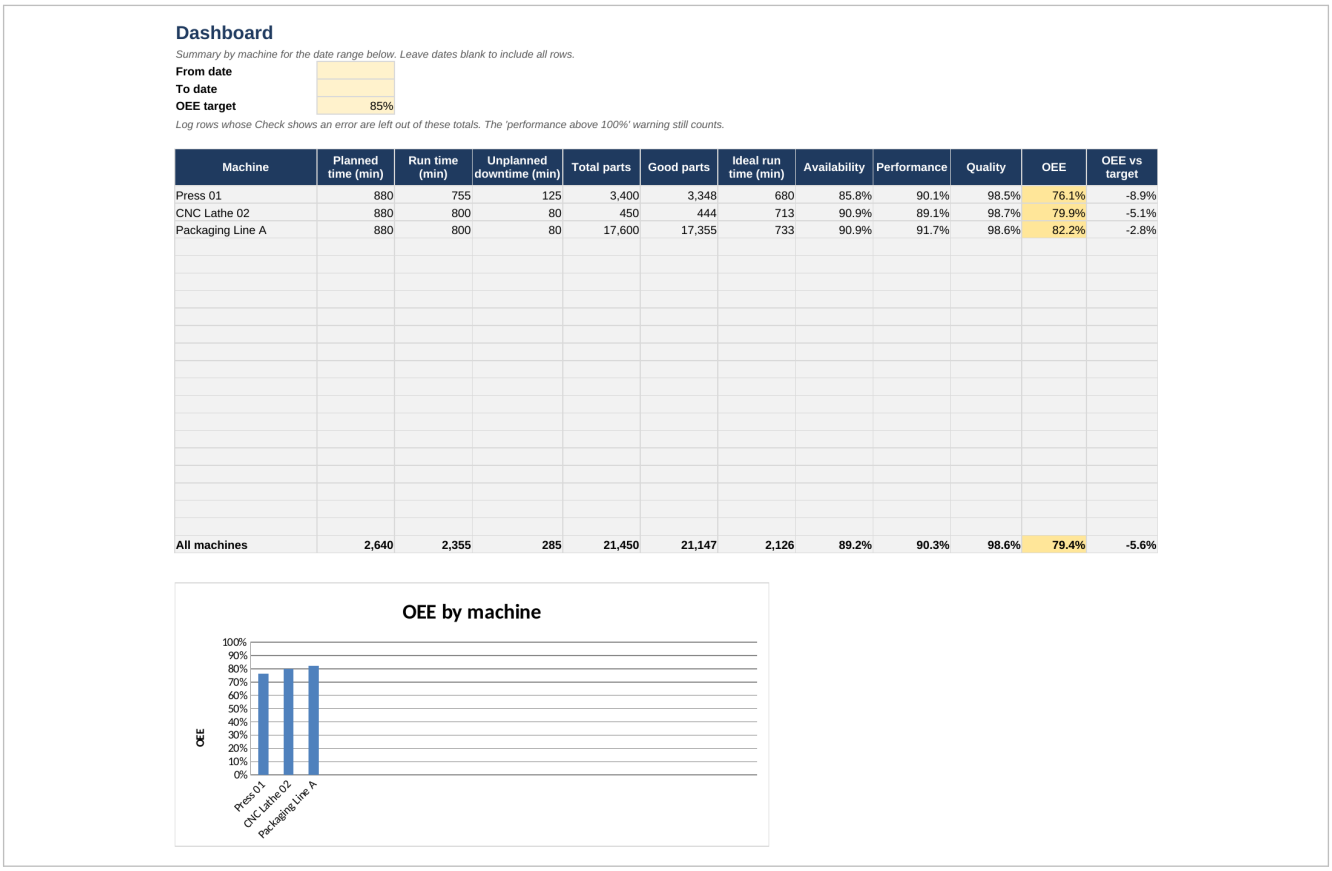
Production Log with the example rows. Yellow columns are typed in; grey columns calculate.

What the Check column means

Message	What to do
Machine not in Setup	The machine name isn't on the Setup sheet, or is spelled differently. Pick it from the drop-down, or add it to Setup.
Set an ideal cycle time in Setup	The machine has no ideal cycle time (or 0) on the Setup sheet.
Enter a date	The Date cell is empty or isn't a real date. Undated rows can't be placed in a date range, so they are left out.
Enter numbers: shift length and total parts are required	A required cell is empty, or a cell contains text instead of a number.
Values must be 0 or more (shift length above 0)	A negative number, or a shift length of 0.
Rejects > total parts	Rejected parts must be part of the total. Fix one of the two numbers.
Planned stops >= shift length	Planned stops leave no time to run. Check both numbers.
Downtime > planned time	Unplanned downtime is longer than shift length minus planned stops. Check the three time columns.
Check: performance above 100%	A warning only; the row still counts. The machine made parts faster than its ideal cycle time allows. Usually the cycle time in Setup is too slow, or the count is wrong.
OK	Nothing obviously wrong.

Step 3: Read the Dashboard

1. Go to the **Dashboard** sheet.
2. Optional: enter a **From date** and/or **To date**. Leave them blank to include every row. Both dates are included in the range.
3. Set your **OEE target** (85% is filled in). OEE cells turn **green** at or above target, **amber** from 60% up to the target, and **red** below 60%. Red always wins: below 60% is red even if your target is lower than 60%. **OEE vs target** shows the gap.
4. Each row adds up times and part counts first, then calculates the percentages, so a long shift counts more than a short one. **All machines** OEE is total ideal time for good parts divided by total planned production time, which stays correct when machines have different cycle times. Its **Quality** is weighted by ideal time (ideal time for good parts ÷ ideal time for all parts), so Availability × Performance × Quality still equals OEE.



Dashboard

Summary by machine for the date range below. Leave dates blank to include all rows.

From date
To date
OEE target 85%

Log rows whose Check shows an error are left out of these totals. The 'performance above 100%' warning still counts.

Machine	Planned time (min)	Run time (min)	Unplanned downtime (min)	Total parts	Good parts	Ideal run time (min)	Availability	Performance	Quality	OEE	OEE vs target
Press 01	880	755	125	3,400	3,348	680	85.8%	90.1%	98.5%	76.1%	-8.9%
CNC Lathe 02	880	800	80	450	444	713	90.9%	89.1%	98.7%	79.9%	-5.1%
Packaging Line A	880	800	80	17,600	17,355	733	90.9%	91.7%	98.6%	82.2%	-2.8%
All machines	**2,640**	**2,355**	**285**	**21,450**	**21,147**	**2,126**	**89.2%**	**90.3%**	**98.6%**	**79.4%**	**-5.6%**

Dashboard: totals and OEE by machine, the all-machines total, and the OEE chart.

How the numbers are calculated

Measure	Formula
Planned production time	Shift length - planned stops
Run time	Planned production time - unplanned downtime
Availability	Run time ÷ planned production time
Performance	(Ideal cycle time × total parts) ÷ run time
Quality	Good parts ÷ total parts (good parts = total - rejected)
OEE	Availability × Performance × Quality = (ideal cycle time × good parts) ÷ planned production time

85% OEE is a commonly cited "world-class" benchmark. Many plants start well below it; improving against your own baseline matters more.

Questions

The log is full (500 rows). What now? Save the workbook as a new file for the next period (for example, one per quarter), clear the yellow cells in the log, and keep going.

Can I add more than 20 machines? Not in this version. Message us if you need more.

Can I rename the sheets or move columns? Please don't. The formulas refer to them by position.

Need help?

Message us through Etsy (the **Message seller** button on our shop or your order). Tell us which file, which sheet, and what you typed. We answer questions about using the templates as described in this guide; we can't do custom changes or set-up work.

These templates were designed with help from AI tools and checked by the seller.